

YEAR 9 DISEASES

➤ TOPIC BOOKLET ⇐



Understand how diseases are caused



Explore the immune system



Investigate pathogens and how they spread



Learn about prevention, treatment and healthy lifestyles



Name: _____



Class: _____



Teacher: _____



How this booklet is organised

This booklet works through all four sub-topics of the Year 9 Diseases topic from the NSW Junior Science syllabus. Each dot point has its own section, followed by a question box with Comprehension, Analysis and Research/Extension questions. Where a practical or research activity is helpful, one is included after the relevant dot point.

Syllabus coverage:

- 2.1 Homeostasis – dot points 2.1.1 to 2.1.5
- 2.2 Infectious and non-infectious diseases – dot points 2.2.1 to 2.2.7
- 2.3 Disease control and prevention – dot points 2.3.1 to 2.3.4
- 2.4 Disease – in Context – dot point 2.4.1

Key vocabulary introduced in this topic:

Homeostasis – maintaining stable internal conditions.

Stimulus – a detectable change in the environment.

Receptor – a structure that detects a stimulus.

Effector – a muscle or gland that produces a response.

Feedback loop – a system that uses the result of a change to control the original change.

Pathogen – an organism that causes disease (bacterium, virus, fungus, protozoan, parasite).

Epidemic / endemic / pandemic – increasing scales of disease spread.

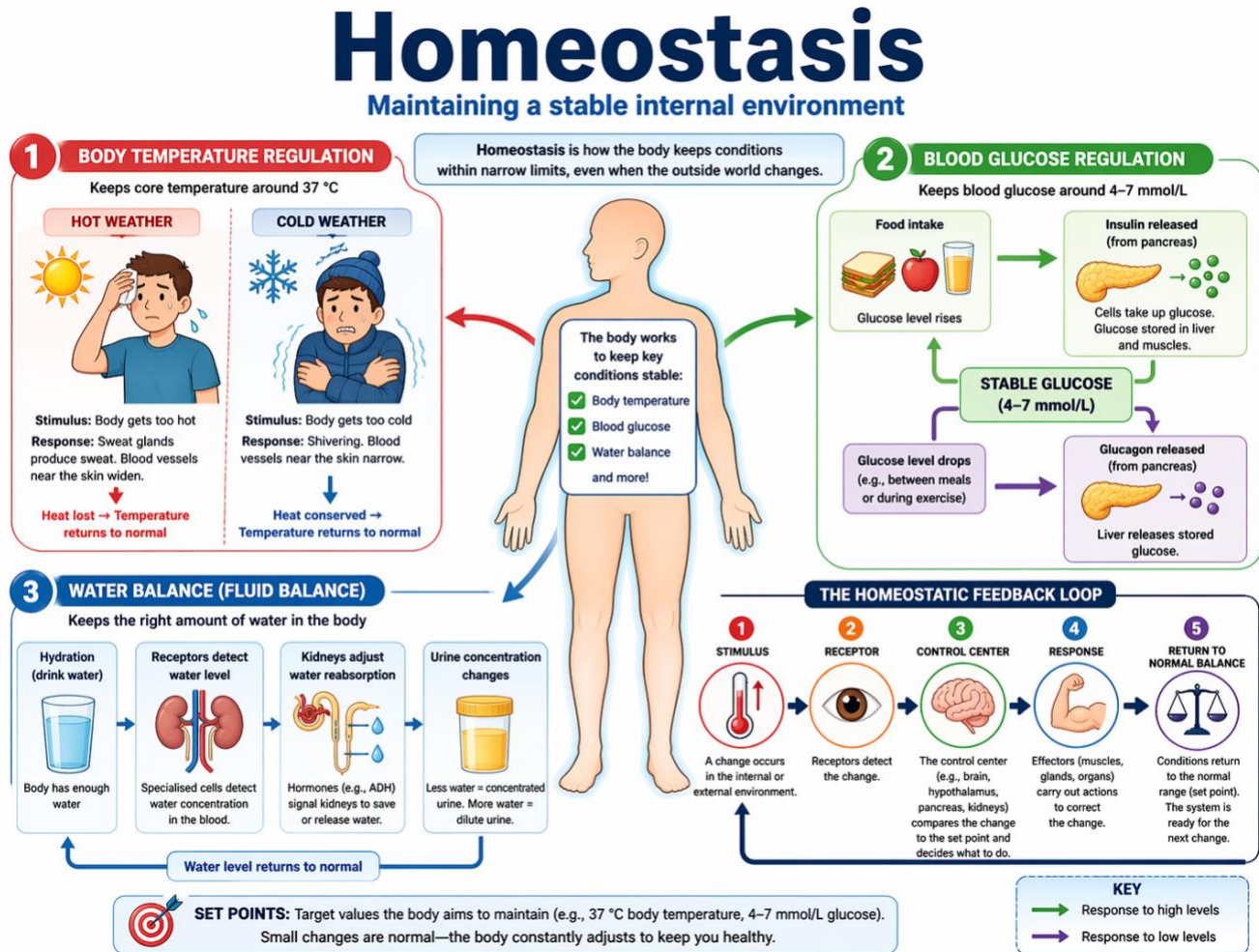
Immune system – the body's defence system against pathogens.

Antibody – a protein made by B-cells that binds specifically to an antigen.

Vaccination – the deliberate exposure of the immune system to a harmless form of a pathogen to create immunity.

2.1 Homeostasis

All living organisms must keep their internal conditions within a narrow range in order to survive. Temperature, water, blood sugar, blood pressure, oxygen, pH, salt balance: each of these is actively kept close to a 'set point' by the body. This section looks at how the body senses change, what it does about it, and which two control systems carry the signals.



2.1.1 Identify the importance of maintaining stable internal conditions in the body

Why must internal conditions stay stable?

The cells of the human body are delicate chemical factories. The enzymes that drive metabolism only work within a narrow range of temperature and pH; nerve cells only fire correctly if the concentration of ions around them is right; the brain demands a steady supply of glucose. If any of these conditions drift too far from their set point, cells stop working and the organism can become very unwell, very quickly.

The process of keeping internal conditions stable is called homeostasis (from the Greek 'homios' meaning same, and 'stasis' meaning standing still). It is important to realise that 'stable' does not mean 'never changing'; conditions always fluctuate a little. Homeostasis means the fluctuations stay within a safe range.

Key variables controlled by homeostasis

Variable	Approximate set point	What happens if it drifts too far
Core body temperature	~37 °C (35.5 – 37.5 °C normal range)	Below 35 °C: hypothermia; above 40 °C: enzymes denature, seizures
Blood glucose	~4 – 8 mmol/L	Low: confusion, coma (hypoglycaemia); high: thirst, organ damage over time (hyperglycaemia)
Blood pH	7.35 – 7.45	Small changes disrupt enzymes and nerve firing; life-threatening below 6.9 or above 7.8
Water / salt balance	~60 % of body mass is water	Dehydration: dizziness, kidney damage; over-hydration: dilutes sodium, can cause seizures
Oxygen and CO ₂ in blood	O ₂ ≥ 95 % saturation	Low O ₂ : cells cannot respire; high CO ₂ : blood becomes acidic

The big idea: every organ in the body has a job to do, but they are all working together to keep the internal environment constant enough that cells can survive. When this system breaks down, we call it a disease.

Questions – 2.1.1

Comprehension Questions

1. Define the term homeostasis.
2. List four variables that the body keeps within a stable range.
3. What is the normal set point for core body temperature?
4. Why do enzymes need a stable internal temperature and pH?

Analysis Questions

1. Explain why 'stable' does not mean 'unchanging' when describing homeostasis.
2. A marathon runner's core temperature rises to 38.5 °C during a race but returns to 37 °C by the next morning. Does this count as homeostasis? Justify your answer.
3. Suggest one reason why blood pH is controlled even more tightly than body temperature.

Research / Extension Questions

1. Research one disease that occurs when homeostasis fails (e.g. Type 1 diabetes, Addison's disease, heat stroke). Describe the variable that is no longer being controlled.
2. Explain what happens to the human body in terms of homeostasis when a person experiences severe dehydration.

2.1.2 Investigate examples of an organism's observable response to a stimulus

A stimulus is any detectable change in the environment. An organism's response is anything observable it does as a result. Both plants and animals respond to stimuli, though plants do so more slowly than animals.

Stimulus	Receptor	Observable response
Bright light	Photoreceptors in the retina	Pupil constricts
Loud sound	Hair cells in the cochlea (ear)	Head turns towards source; startle response
Cold air on skin	Thermoreceptors in the skin	Shivering; goose bumps; blood vessels constrict
Painful pinprick	Pain receptors (nociceptors)	Hand pulls away (reflex arc)
Low blood glucose	Alpha cells in pancreas	Glucagon released; liver releases glucose
Light on leaves (plant)	Phytochromes in leaf cells	Stem grows towards light (phototropism)
Touch on Venus flytrap	Trigger hairs on leaf	Leaf snaps shut (thigmotropism)

Reflexes vs. conscious responses

Some responses are reflexes: fast, automatic, and handled by the spinal cord without the brain needing to think. Touching something sharp and pulling your hand away is one example. Other responses are conscious: you see a ball heading towards you; your brain decides to catch it. Both are responses to a stimulus, but they use different pathways.

Practical 2.1.2 – Measuring reaction time and reflex responses

Aim

To measure observable responses to a stimulus and compare reflex and conscious responses.

Materials

- 30 cm ruler
- Stopwatch
- Torch (small)
- Partner

Part A – Ruler drop (conscious response to a visual stimulus)

1. Partner holds a ruler vertically, 0 cm at the bottom, with the 0 cm mark level with your thumb and finger.
2. Partner releases the ruler without warning. Catch it as fast as you can.
3. Record the distance the ruler fell (cm).
4. Repeat 10 times. Record the mean distance.
5. Swap roles.

Part B – Pupil reflex (reflex response to light)

1. Have your partner sit in a well-lit room for 30 seconds.
2. Observe the size of the pupils.
3. Ask them to close their eyes for 1 minute, then open them while you observe.
4. Describe any change to the pupil size.
5. Shine the torch briefly to one side of one eye (NOT directly into the eye). Observe.

Questions

1. In Part A, what was the stimulus and what was the response?
2. In Part B, what was the stimulus and what was the response?
3. Which response was faster, and why?
4. Which of your responses involved the brain?
5. Identify one controlled variable in Part A and explain why it needs to be controlled.

Questions – 2.1.2

Comprehension Questions

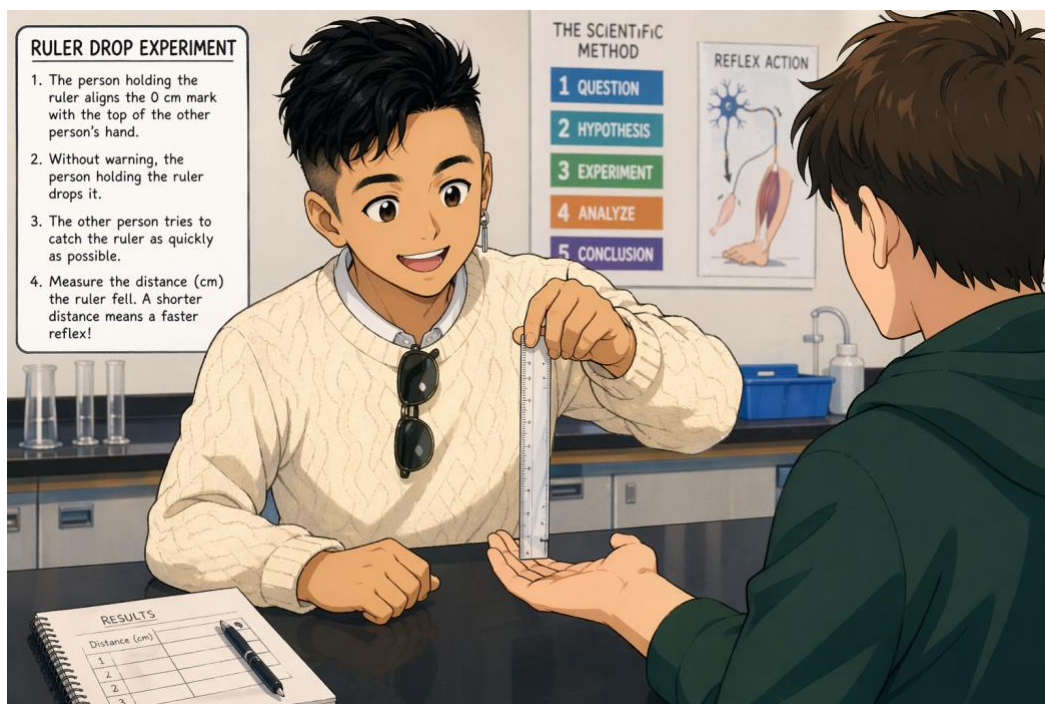
1. Define stimulus and response in your own words.
2. Give two examples of a stimulus that a plant can respond to.
3. What is a reflex?
4. Name a receptor found in the human skin and the stimulus it detects.

Analysis Questions

1. Compare a reflex response and a conscious response. Which is faster and why?
2. A jellyfish has no brain, yet it still responds to touch. Explain how this is possible.
3. Design a fair test that would let you compare the reaction time of two students to a visual stimulus. Identify the independent, dependent and controlled variables.

Research / Extension Questions

1. Research how Venus flytraps detect when an insect has landed on them and how they 'decide' whether to close.
2. Investigate one way that elite athletes train to shorten their reaction time.



2.1.3 Identify the role of feedback loops in maintaining homeostasis

A feedback loop is a system that uses information about the result of a change to adjust its behaviour. The body uses feedback loops constantly; in fact, almost every homeostatic process relies on one.

A Negative Feedback Loop Maintains Homeostasis

Homeostasis keeps internal conditions stable by returning the body to its set point.

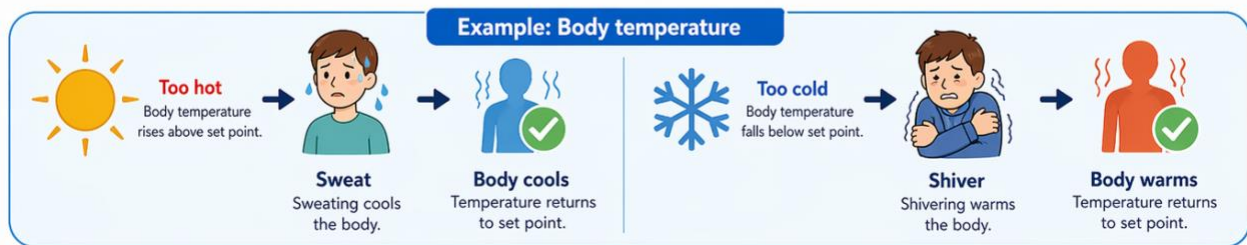
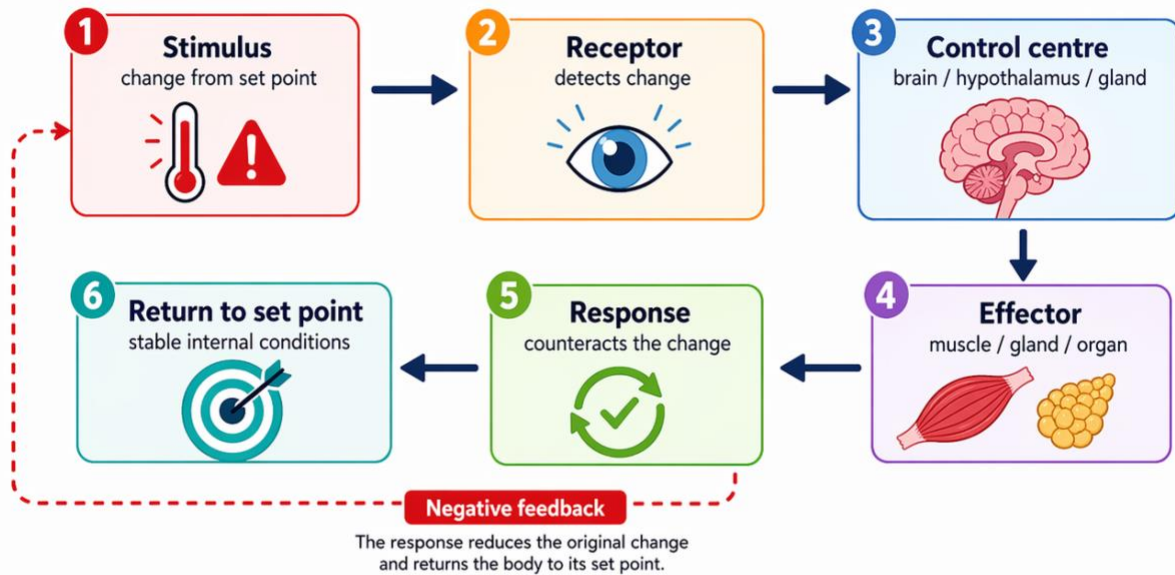


Figure 2.1 – The five stages of a negative feedback loop.

Negative feedback

In a negative feedback loop, a change away from the set point triggers a response that reverses the change. 'Negative' here does not mean 'bad'; it means 'opposing'. Negative feedback is the dominant system in the human body because it keeps variables stable. Examples include temperature regulation, blood glucose control, blood pressure, and water balance.

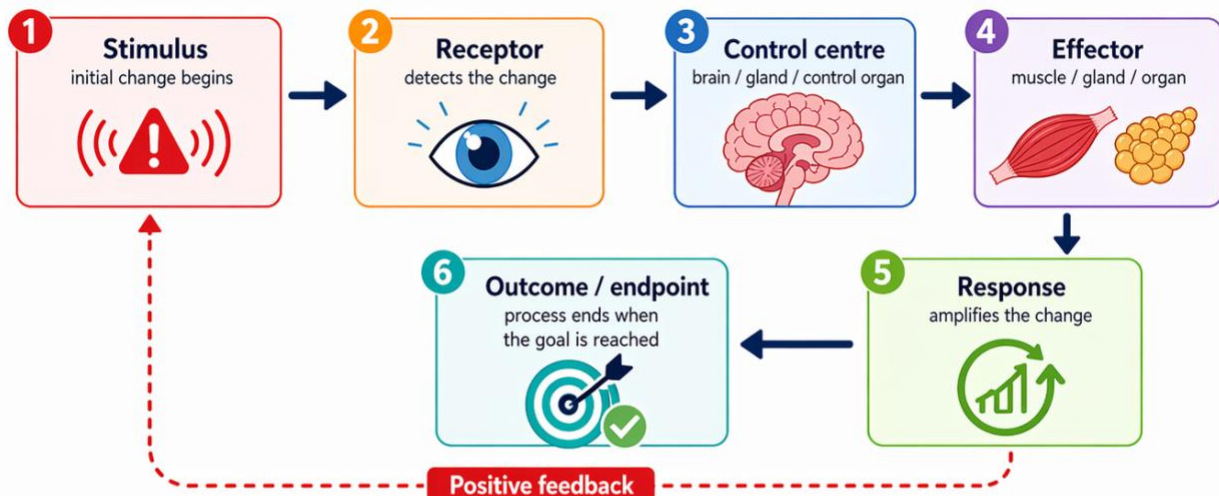
Positive feedback

In a positive feedback loop, a change away from the set point triggers a response that increases the change. Positive feedback is less common in the body and is usually used when a process needs to reach completion quickly. Examples include blood clotting (each stage of clotting releases chemicals that speed up the next stage) and childbirth (contractions cause the release of oxytocin, which causes stronger contractions).

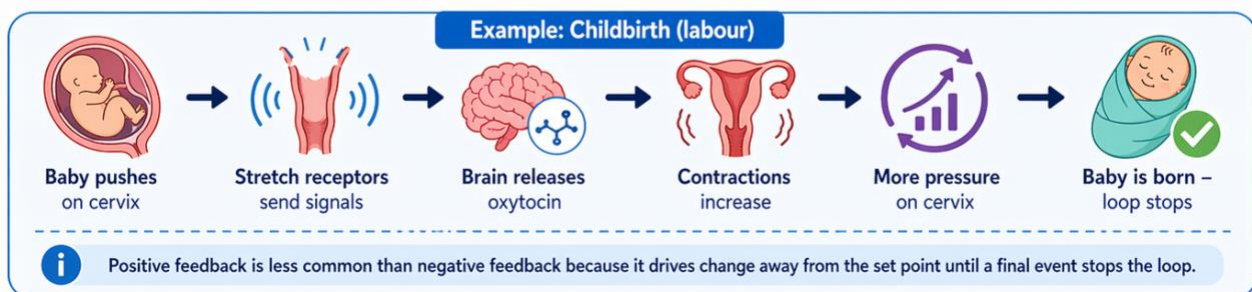
Feature	Negative feedback	Positive feedback
Effect on change	Reverses the change	Amplifies the change
Purpose	Restore stable conditions	Drive a process to completion
Common in body?	Very common	Rare
Example	Temperature regulation	Blood clotting, childbirth

A Positive Feedback Loop Amplifies Change

Positive feedback increases the original change until a specific outcome is reached.



The response strengthens the original stimulus, causing more change until the endpoint is reached.



Positive feedback is less common than negative feedback because it drives change away from the set point until a final event stops the loop.

Questions – 2.1.3

Comprehension Questions

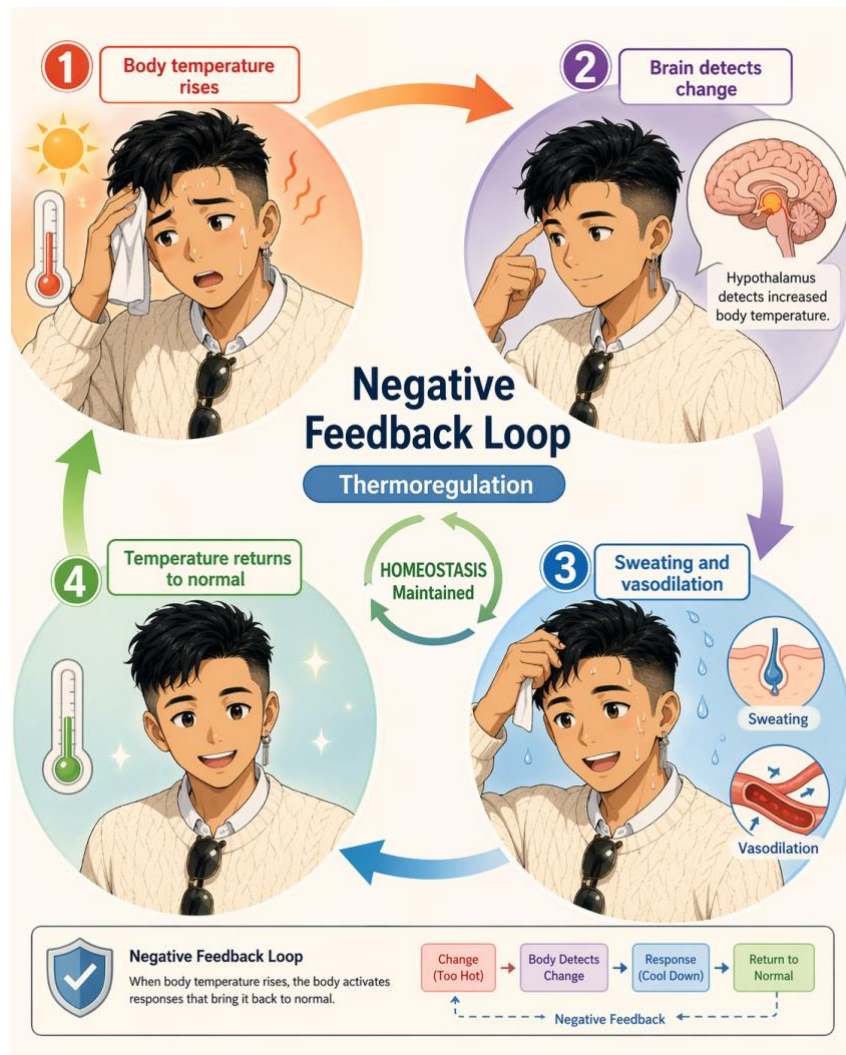
1. What is a feedback loop?
2. Name the five stages of a negative feedback loop.
3. Does 'negative feedback' mean 'bad'? Explain.
4. Give one example of positive feedback in the human body.

Analysis Questions

1. Explain why negative feedback is more common in the body than positive feedback.
2. A thermostat in a house keeps the temperature at 22 °C. Identify the stimulus, receptor, control centre, effector and response in this system.
3. Why would it be dangerous if body temperature was controlled by positive feedback?

Research / Extension Questions

1. Research how the hormone oxytocin causes positive feedback during childbirth.
2. Investigate the term 'allostasis' and explain how it differs from homeostasis.



2.1.4 Compare and contrast the responses of the nervous and endocrine systems

The body has two coordinating systems that carry signals from receptors to effectors: the nervous system and the endocrine system. They work together, but they are very different.

Two control systems, different speeds and durations

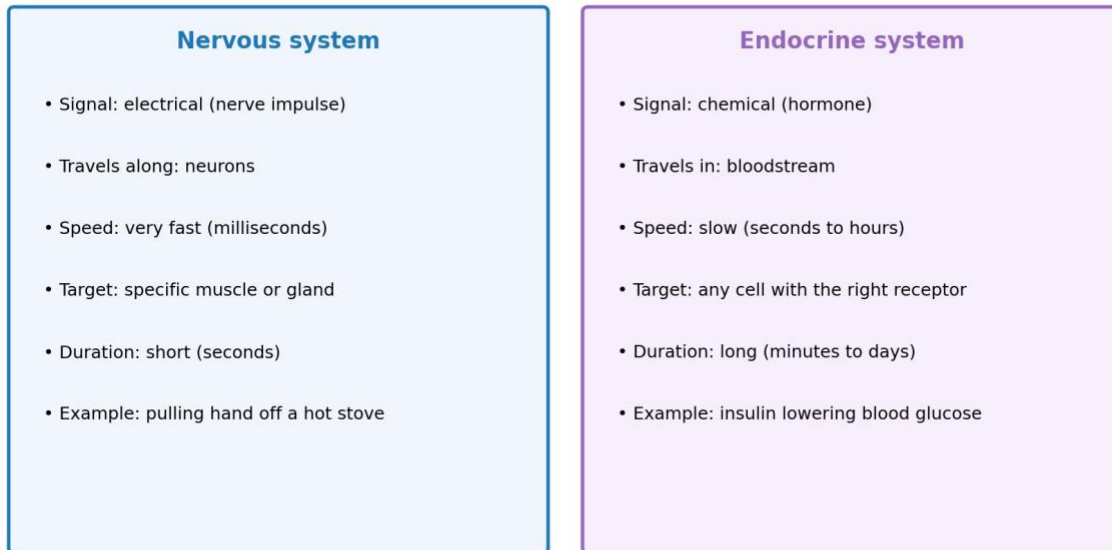


Figure 2.2 – The nervous and endocrine systems compared.

The nervous system

The nervous system uses electrical impulses that travel along nerve cells (neurons). Because the signal is electrical and runs along a dedicated 'wire', it is extremely fast, measured in milliseconds. The signal only reaches the specific muscle or gland that the nerve connects to. Nervous responses are usually very short-lived: once the impulse stops, the response stops.

The endocrine system

The endocrine system uses chemical messengers called hormones. Glands release hormones into the bloodstream, which carries them to every cell in the body. Only cells with matching receptors respond, but because the hormone is dissolved in the blood, the signal is much slower: seconds, minutes or even hours. Endocrine responses tend to last longer, because hormones circulate for a while before being broken down.

Feature	Nervous system	Endocrine system
Type of signal	Electrical (nerve impulse)	Chemical (hormone)
Pathway	Along neurons	Through the bloodstream
Speed of action	Very fast (milliseconds)	Slower (seconds to hours)
Duration of effect	Short (seconds)	Long (minutes to days)
Specificity	Only the cells the nerve ends at	Any cell with the correct receptor
Typical examples	Reflexes, movement, vision, pain	Growth, metabolism, puberty, blood glucose
Main control centre	Brain and spinal cord	Pituitary gland (and hypothalamus)

Questions – 2.1.4

Comprehension Questions

1. What kind of signal does the nervous system use?
2. What kind of signal does the endocrine system use?
3. Which system acts faster – nervous or endocrine?
4. Which system usually produces a longer-lasting response?

Analysis Questions

1. Explain why hormones can only affect certain cells, even though they travel everywhere in the bloodstream.
2. Imagine the endocrine system tried to carry out a reflex response. Why would this not work?
3. Construct a Venn diagram that compares the nervous system and endocrine system.

Research / Extension Questions

1. Research three hormones (other than insulin) and record what gland releases them and what they do.
2. Investigate the role of the hypothalamus in linking the nervous system and the endocrine system.

Nervous System and Endocrine System

Two body systems that help your body communicate and respond!

NERVOUS SYSTEM

The body's **quick** communication network.

- BRAIN**: Control center
- SPINAL CORD**: Information highway
- NERVES**: Carry signals throughout the body

- ⚡ Uses electrical impulses
- ⌚ Fast response
- ⌚ Short-lasting effect

EXAMPLE: TOUCHING SOMETHING HOT

1. You touch something hot.
2. Nerves send a signal to the spinal cord and brain.
3. You pull your hand away—**fast!**

ENDOCRINE SYSTEM

The body's **chemical** messenger system.

- PITUITARY GLAND**: "Master gland" — controls other glands
- THYROID GLAND**: Helps control metabolism
- ADRENAL GLANDS**: Help with stress response
- PANCREAS**: Helps control blood sugar
- REPRODUCTIVE GLANDS**: Involved in puberty and reproduction

- 💧 Uses hormones in the blood
- ⌚ Slower response
- ⌚ Longer-lasting effect

EXAMPLE: PUBERTY (GROWTH AND CHANGES)

1. Hormones are released by glands.
2. Hormones travel in the blood to different parts of the body.
3. Changes happen over time, like growth and voice deepening.

Nervous System vs Endocrine System

Nervous System		Endocrine System
Electrical impulses (nerve signals)	💬 Message type	Chemical hormones in the blood
Very fast (milliseconds to seconds)	⌚ Speed	Slower (seconds to hours or days)
Short-lasting (seconds to minutes)	⌚ Effect duration	Long-lasting (hours to years)

Remember: The nervous system is like the **BODY'S INTERNET**.
The endocrine system is like the **BODY'S MAIL SERVICE**.

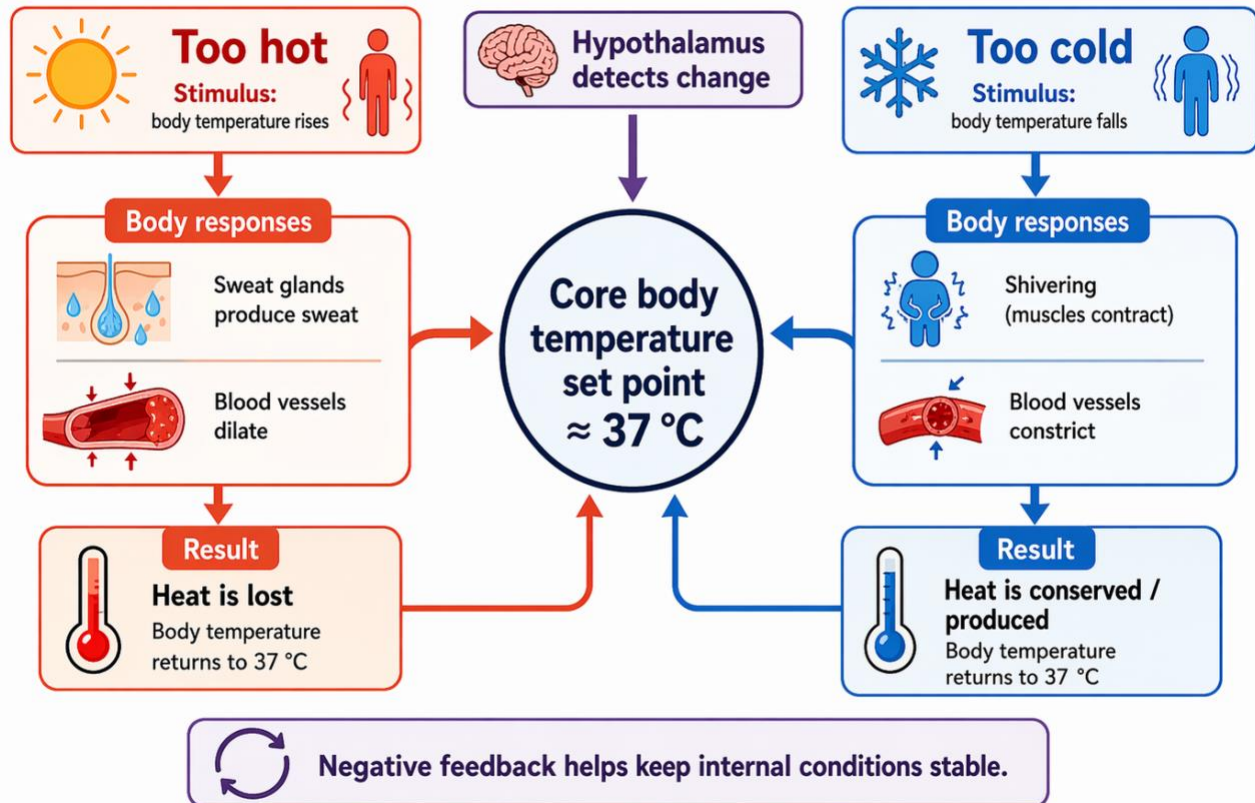
Together, they help you think, feel, grow and stay healthy!

2.1.5 Describe how the nervous and endocrine systems coordinate the body's response to stimuli

The nervous and endocrine systems rarely work alone. For most real-life stimuli, both systems are involved: the nervous system handles the fast initial response and the endocrine system handles the longer, sustained response.

Worked example: thermoregulation (temperature control)

Maintaining Thermoregulation in Humans



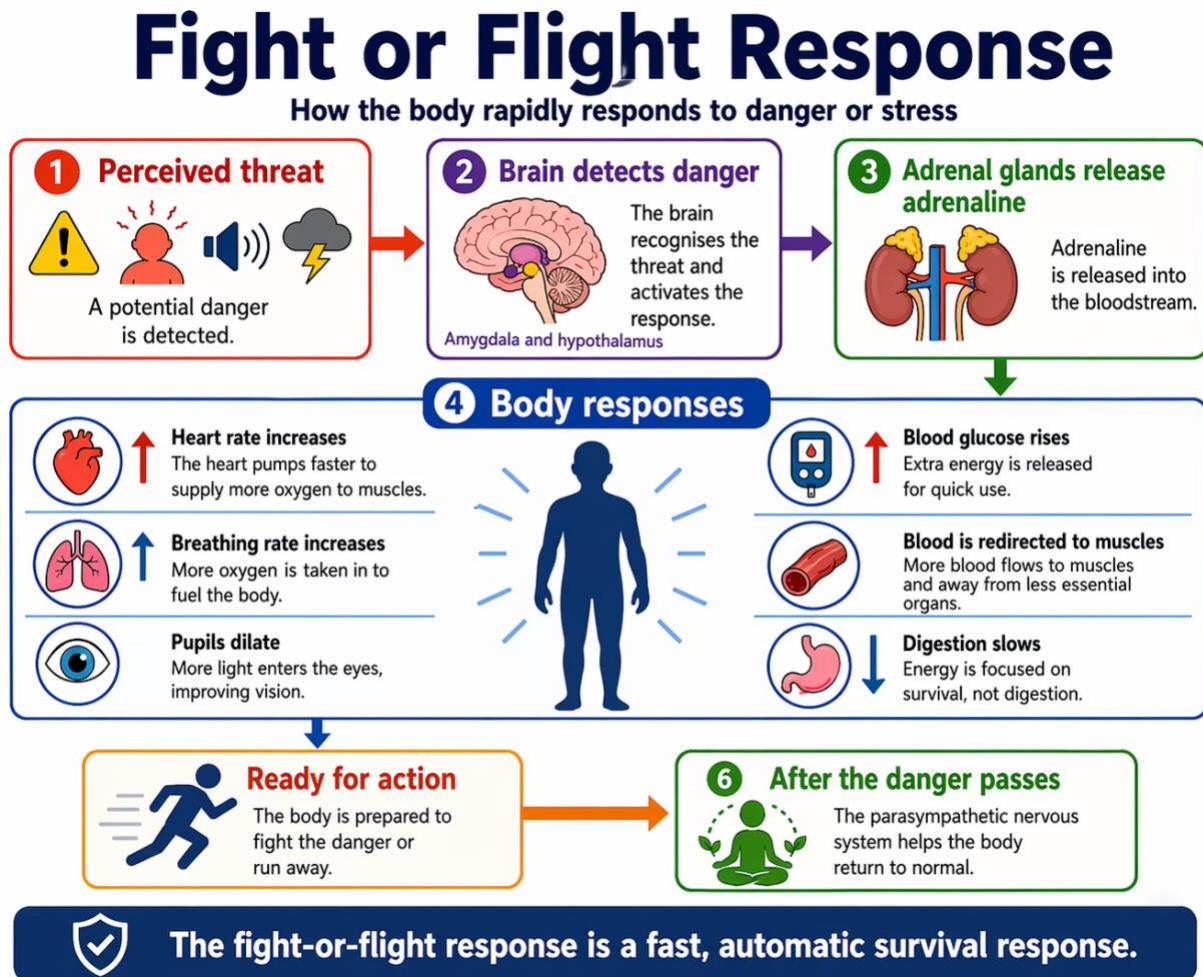
Thermoregulation helps the body respond to being too hot or too cold.

Figure 2.3 – How the body responds to being too hot or too cold.

When you go outside on a cold day, thermoreceptors in the skin detect the drop in temperature. Nerves carry the signal to the hypothalamus in the brain (the control centre). The hypothalamus uses the nervous system to trigger a fast response: skeletal muscles shiver, and blood vessels in the skin constrict to keep blood warm. At the same time, the hypothalamus uses the endocrine system to stimulate the thyroid gland to release thyroxine, which raises the metabolic rate of cells, generating more heat over the long term.

Worked example: responding to danger (fight-or-flight)

Imagine a car suddenly swerves towards you. Your eyes send signals through the nervous system to your brain. Within a fraction of a second, the brain activates motor neurons so your legs leap out of the way, which is a nervous response. At the same time, the brain signals the adrenal glands to release adrenaline into the blood. Adrenaline causes the heart to beat faster, the breathing rate to increase, and blood to be diverted to the muscles. That's the endocrine response, kicking in a few seconds later but lasting much longer. Together the two systems give you an immediate escape and a sustained energy boost.



Worked example: blood glucose control

After a meal, blood glucose rises. The pancreas (endocrine gland) releases the hormone insulin, which tells body cells to absorb glucose and the liver to store glucose as glycogen. Between meals, blood glucose drops. The pancreas releases a different hormone, glucagon, which tells the liver to break down glycogen and release glucose. The two hormones act as a pair of opposites, holding blood glucose inside a safe range. This is almost entirely an endocrine response, though the nervous system is involved in detecting hunger.

Questions – 2.1.5

Comprehension Questions

1. Which part of the brain is the control centre for temperature regulation?
2. Name two effectors that help cool the body when it is too hot.
3. Which hormone is released by the pancreas when blood glucose is high?
4. Which hormone is released by the adrenal glands during fight-or-flight?

Analysis Questions

1. Describe how the nervous and endocrine systems work together during the fight-or-flight response.
2. Explain why an endocrine response alone would not be fast enough to keep you safe from a swerving car.
3. A person has a damaged pancreas and cannot produce insulin. Predict the effect this will have on their blood glucose after a meal.

Research / Extension Questions

1. Research Type 1 diabetes. Explain which part of the homeostasis pathway for blood glucose has failed.
2. Investigate how the body responds to water loss (dehydration). Which hormone is involved, and which organ is the main effector?



2.2 Infectious and non-infectious diseases

A disease is any condition that stops the body from functioning normally. Some diseases can be passed between people (infectious); others cannot (non-infectious). Both are responsible for major health problems in Australia and across the world, but they are caused, spread and prevented in different ways.

2.2.1 Distinguish between infectious and non-infectious diseases

Infectious diseases

Infectious diseases are caused by a pathogen, which is a living organism (or a virus) that has entered the body and is causing harm. Because the pathogen can move to another person, infectious diseases can spread. They are also called communicable or transmissible diseases. Common examples include the common cold, influenza, COVID-19, chickenpox, tuberculosis, and food poisoning.

Non-infectious diseases

Non-infectious diseases are not caused by a pathogen and cannot be passed from one person to another. They are caused by other factors such as genetics, lifestyle, environment or age. Common examples include type 2 diabetes, most cancers, cardiovascular disease, asthma, and dementia. Because they often develop slowly and last a long time, they are also called chronic or non-communicable diseases (NCDs).

Feature	Infectious disease	Non-infectious disease
Cause	Pathogen (bacterium, virus, fungus, protozoan, parasite)	Genetic, lifestyle, environment, age
Contagious?	Yes – can spread between people	No – cannot be passed on
Onset	Often sudden (hours to days)	Often slow (months to years)
Duration	Usually short (days to weeks)	Often long-term or lifelong
Examples	Influenza, COVID-19, measles, tinea	Heart disease, type 2 diabetes, asthma
Main treatment	Antibiotics, antivirals, rest, vaccination	Lifestyle change, ongoing medication, surgery

Questions – 2.2.1

Comprehension Questions

1. Define an infectious disease.
2. Define a non-infectious disease.
3. List two examples of each type of disease.
4. What is another name for a non-infectious disease?

Analysis Questions

1. Classify each of the following as infectious or non-infectious: measles, asthma, malaria, diabetes, chickenpox, cancer, HIV/AIDS, coronary heart disease.
2. A student claims 'all diseases can be caught from someone else'. Explain why this statement is wrong.
3. Explain why infectious diseases often appear suddenly, while non-infectious diseases develop slowly.

Research / Extension Questions

1. Investigate an infectious disease that is now rare in Australia due to improved health care.
2. Research one non-infectious disease that has become more common in Australia over the last 50 years and suggest why.

2.2.2 Identify causes of non-infectious and infectious diseases

Causes of infectious diseases

Every infectious disease has a specific pathogen as its cause. There are five main types of pathogens.

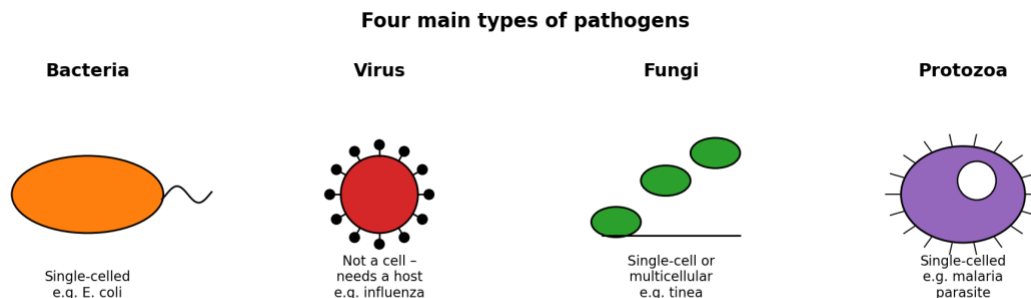


Figure 2.4 – The four main cellular and non-cellular pathogens. Parasitic worms are a fifth group.

Pathogen type	Living / non-living	Examples of disease
Bacteria	Living, single-celled	Tuberculosis, strep throat, food poisoning (Salmonella)
Viruses	Non-living (needs a host cell)	Influenza, COVID-19, measles, HIV
Fungi	Living, uni- or multicellular	Tinea (athlete's foot), thrush, ringworm
Protozoa	Living, single-celled	Malaria, giardia
Parasitic worms	Living, multicellular	Tapeworm, hookworm

Causes of non-infectious diseases

Non-infectious diseases have multiple causes. These are usually grouped into risk factors – conditions that make a person more likely to develop the disease.

Category	Examples of risk factors	Examples of diseases
Genetic / inherited	Faulty gene, family history	Cystic fibrosis, haemophilia, some cancers
Lifestyle	Smoking, poor diet, inactivity, alcohol	Heart disease, type 2 diabetes, lung cancer
Environmental	Air pollution, UV exposure, chemical exposure	Skin cancer, asthma, mesothelioma
Age / developmental	Ageing cells, hormonal changes	Osteoporosis, dementia
Nutritional	Lack of iodine, iron, vitamins, kilojoules	Anaemia, goitre, rickets

Importantly, many non-infectious diseases have more than one cause. For example, type 2 diabetes is influenced by both genetics and lifestyle. This makes them harder to prevent with a single action.

Questions – 2.2.2

Comprehension Questions

1. Name the five main groups of pathogens.
2. Which pathogen group is not made of cells?
3. List three risk factors for non-infectious disease.
4. Give one example of a disease caused by: a bacterium, a virus, a fungus.

Analysis Questions

1. Explain why a virus is said to be 'non-living' outside of a host.
2. Classify the cause of each of the following as genetic, lifestyle, environmental, or pathogen: lung cancer in a smoker, malaria, skin cancer, cystic fibrosis, strep throat.
3. Why is it harder to prevent a non-infectious disease than an infectious disease?

Research / Extension Questions

1. Research one bacterial disease and one viral disease. Compare how each can be treated.
2. Investigate why antibiotics only work on bacterial infections, not on viral infections.



2.2.3 Compare the features and incidences of epidemics, endemics and pandemics

Disease spread is described using three key terms: endemic, epidemic and pandemic. The word 'incidence' means the number of new cases of a disease reported in a population over a period of time.

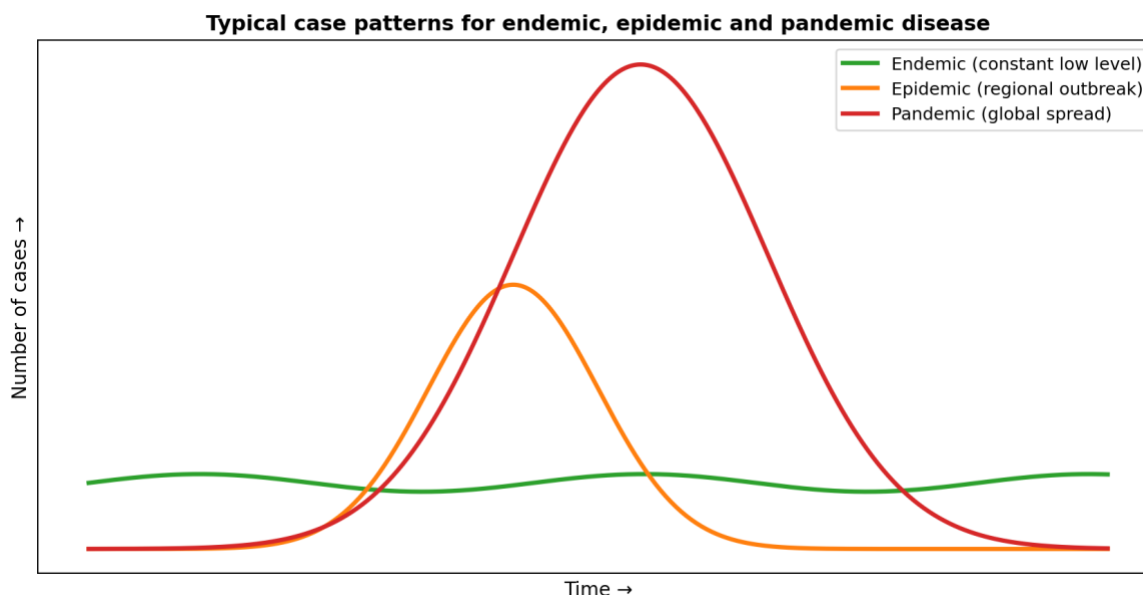


Figure 2.5 – Typical case patterns for endemic, epidemic and pandemic diseases.

Term	What it means	Example
Endemic	A disease that is constantly present in a population or region, usually at a low level.	Malaria in parts of sub-Saharan Africa; dengue fever in northern Queensland.
Epidemic	A sudden, larger-than-expected increase in cases of a disease in a particular region.	Cholera outbreak in Haiti, 2010; measles outbreak in Samoa, 2019.
Pandemic	An epidemic that has spread across several countries or continents, affecting large numbers of people.	COVID-19 (2020–2023); 1918 'Spanish flu'; HIV/AIDS.

The same disease can be described differently depending on where and when you are looking. For example, influenza is endemic in Australia (every year there are some cases), but in a bad winter it can become epidemic (a spike of cases). In 2009, the H1N1 strain of influenza spread worldwide and was declared a pandemic.

Incidence vs prevalence

Incidence = the number of new cases in a period of time.

Prevalence = the total number of cases (old and new) in a population at a given time.

A disease can have a low incidence but a high prevalence if it lasts a long time (for example, HIV).

Questions – 2.2.3

Comprehension Questions

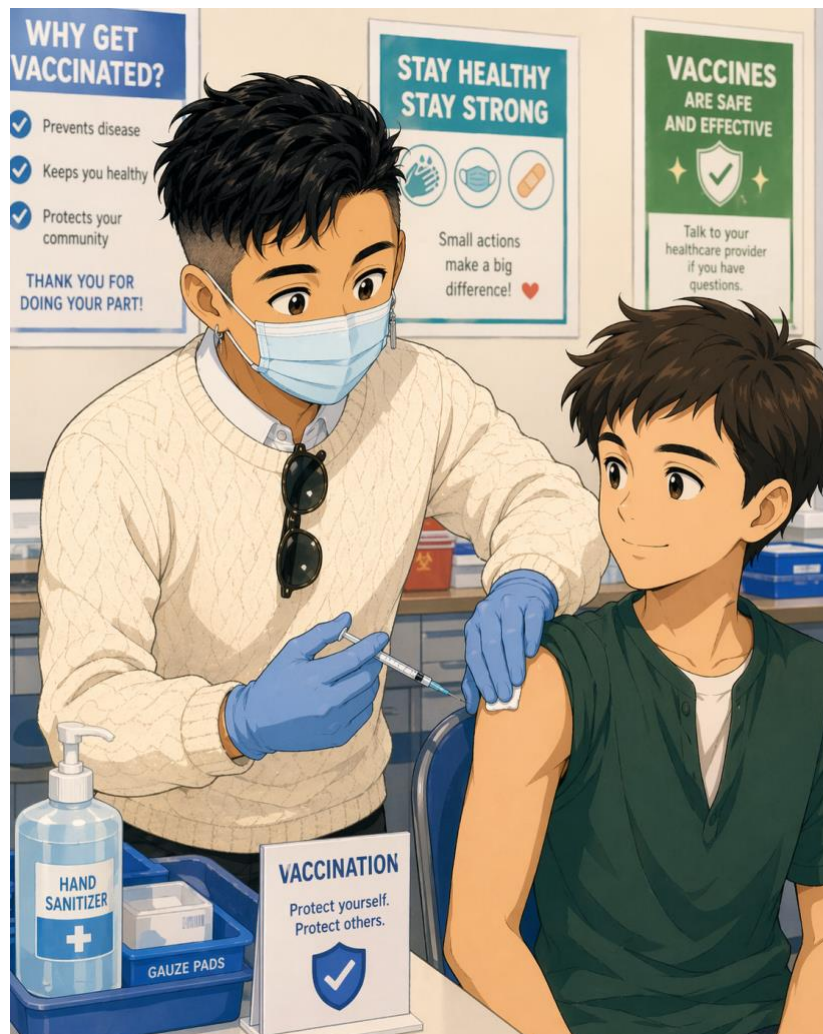
1. Define endemic, epidemic and pandemic.
2. Give one example of a disease that is currently endemic in Australia.
3. What does the term 'incidence' refer to?
4. Is influenza an endemic, epidemic or pandemic disease? Explain.

Analysis Questions

1. Explain why a disease that is endemic in one region may be considered epidemic if it suddenly appears somewhere else.
2. COVID-19 was declared a pandemic in March 2020 by the World Health Organization. Justify the use of the word 'pandemic' based on the criteria in this section.
3. Using Figure 2.5, describe three key differences between the three case curves.

Research / Extension Questions

1. Research one pandemic from history (for example, the 1918 Spanish flu, the 14th-century Black Death, or HIV/AIDS). Record what caused it, how it spread, and approximately how many people died.
2. Investigate why malaria is described as endemic in some parts of the world but not in Australia.



2.2.4 Investigate data relating to a common non-infectious disease affecting Australians today

Non-infectious (chronic) diseases are the biggest cause of death and disability in Australia. According to the Australian Institute of Health and Welfare, chronic diseases are responsible for roughly 9 out of every 10 deaths in Australia. Four of the biggest are cardiovascular disease, cancer, type 2 diabetes and chronic respiratory disease (including asthma and COPD).

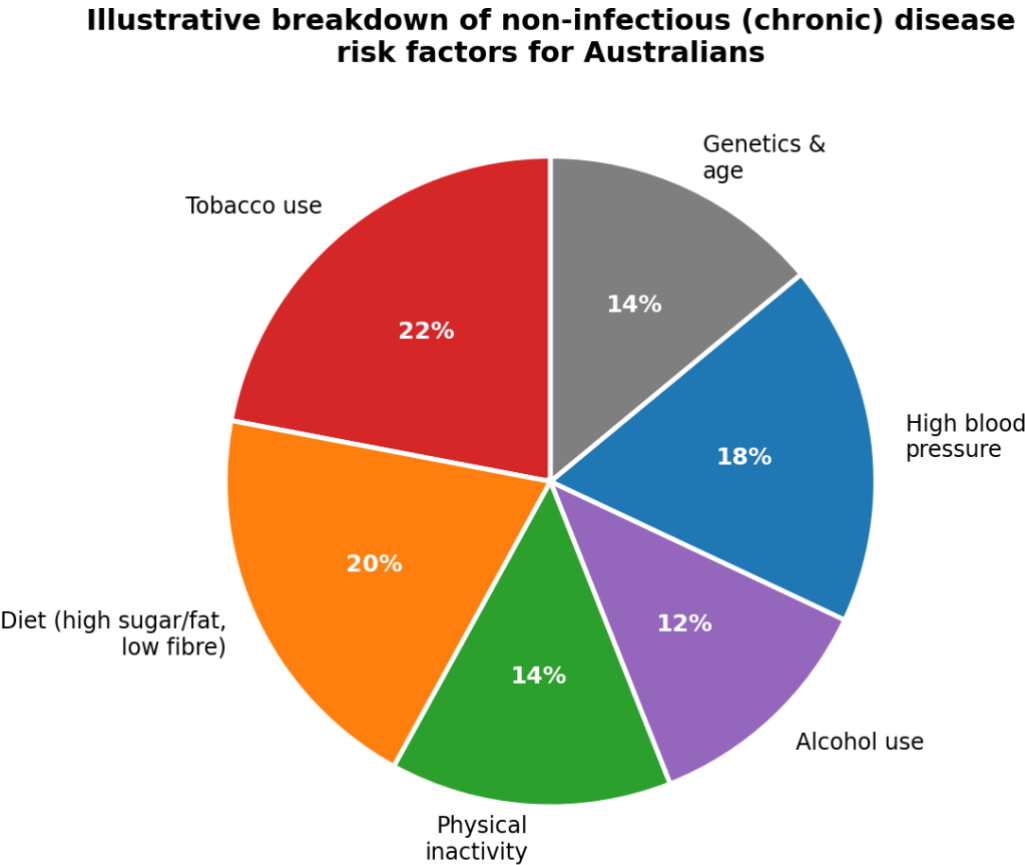


Figure 2.6 – Illustrative breakdown of major risk factors for chronic disease in Australia.

Case study: cardiovascular disease (CVD)

Cardiovascular disease is a group of conditions affecting the heart and blood vessels, including coronary heart disease, stroke, and heart failure. It is the leading single cause of death in Australia.

Statistic (indicative)	Value
Proportion of Australian deaths from cardiovascular disease (approx.)	about 1 in 4
Australians living with cardiovascular disease (approx.)	about 1.2 million
Annual cost to the Australian health system (approx.)	over \$12 billion
Higher-risk age group	over 65
Main modifiable risk factors	smoking, high blood pressure, high cholesterol, obesity, physical inactivity, diet high in saturated fat and salt
Main non-modifiable risk factors	age, sex, family history

Data investigation – chronic disease in Australia

Task

Using a reliable source (such as the Australian Institute of Health and Welfare – aihw.gov.au – or the Australian Bureau of Statistics – abs.gov.au), investigate ONE common non-infectious disease affecting Australians today. Suggested choices: cardiovascular disease, type 2 diabetes, lung cancer, melanoma, or asthma.

Data to collect

1. Approximately how many Australians have the disease, or die from it each year?
2. Which age group is most affected?
3. What are the main risk factors (both modifiable and non-modifiable)?
4. Has the number of cases increased, decreased, or stayed the same over the last 20 years?
5. What can individuals do to reduce their risk?

Presentation

Present your findings as a one-page poster, infographic or short written report. Include at least one graph or table taken from your source, clearly labelled and with the source cited.

Questions – 2.2.4

Comprehension Questions

1. What does 'CVD' stand for?
2. Approximately what proportion of deaths in Australia are from chronic disease?
3. List three modifiable risk factors for cardiovascular disease.
4. Name a non-modifiable risk factor for heart disease.

Analysis Questions

1. Explain the difference between a modifiable and a non-modifiable risk factor.
2. Two students have the same family history of heart disease. Student A exercises regularly and does not smoke. Student B is inactive and smokes. Predict whose risk is higher and explain why.
3. Use Figure 2.6 to identify the largest group of risk factors for chronic disease. Suggest one policy the Australian government could introduce to target it.

Research / Extension Questions

1. Research how Australia compares to other developed countries for the rate of cardiovascular disease.
2. Investigate one prevention program currently run in Australia (for example, the Heart Foundation's 'Heart Week' or the 'LiveLighter' campaign).

2.2.5 Use modelling to investigate how infectious diseases can be spread

Scientists use models to investigate things that are hard, dangerous or too slow to study directly. Modelling the spread of an infectious disease lets us see how quickly a disease can move through a population, and how different behaviours change that spread.

The chain of infection - break any link to stop the spread

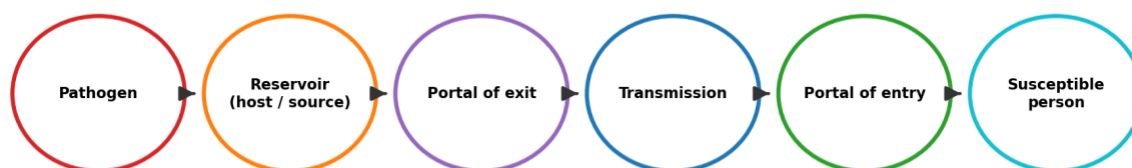


Figure 2.7 – The chain of infection. Every link must be in place for transmission to occur.

Modes of transmission

Mode of transmission	How it works	Example disease
Droplet (cough/sneeze)	Infected droplets land on another person's mucous membranes	Influenza, COVID-19
Airborne	Tiny particles remain in the air for minutes to hours	Tuberculosis, measles
Direct contact	Skin-to-skin, or contact with open wounds or bodily fluids	Cold sores, Ebola
Indirect contact (fomite)	Pathogen survives on a surface that is then touched	Gastro viruses, influenza
Food and water	Eating or drinking contaminated food or water	Salmonella, cholera
Vector-borne	An animal (often an insect) carries the pathogen	Malaria (mosquito), dengue (mosquito)
Mother to baby	Across the placenta, during birth, or through breast milk	HIV, rubella

Practical 2.2.5 – Modelling the spread of an infectious disease

Aim

To model how an infectious disease spreads through a population and to investigate how different factors change the rate of spread.

Materials

- 1 plastic cup per student, three-quarters full of water
- 1 cup secretly contains a few drops of sodium hydroxide (NaOH) – this is the 'infected' cup
- A pipette per student
- Universal indicator / phenolphthalein (added at the end for testing)

Method

1. Each student receives a cup. Do NOT look inside your own or anyone else's cup.
2. On the teacher's signal, use the pipette to transfer a small amount of liquid from your cup to ONE other student's cup, and let them do the same to yours.
3. Repeat for 3 rounds with different partners each time. Record who you 'contacted' each round.
4. After 3 rounds, the teacher adds a few drops of indicator to each cup.
5. Pink/purple = 'infected'. Record who is now infected.

Safety

Sodium hydroxide is corrosive – wear safety glasses and do not drink from any cup. Your teacher will set up the 'infected' cup and dispose of all materials at the end.

Extension investigation

1. Repeat the model but have 25 % of students 'vaccinated' (they do not exchange liquid). What happens?
2. Repeat but have all students wash their pipette in water between rounds ('hand-washing'). What happens?
3. Use the class record to work backwards and identify who was 'patient zero'.

Questions

1. What real-world feature of infectious disease spread does this model represent well?
2. What are two limitations of this model?
3. Explain why 'vaccinating' 25 % of students reduced the spread, even for students who were not vaccinated.
4. How does this model relate to the chain of infection in Figure 2.7?

Questions – 2.2.5

Comprehension Questions

1. List four modes of transmission of infectious diseases.
2. What is a vector? Give one example.
3. What is a fomite?
4. Name the six links in the chain of infection.

Analysis Questions

1. Explain why breaking any one link in the chain of infection can stop a disease spreading.
2. For each of the following diseases, identify the main mode of transmission: influenza, malaria, gastroenteritis, HIV, athlete's foot.
3. Suggest three reasons a model of disease spread is useful for public health planners, even though it is simplified.

Research / Extension Questions

1. Research the term ' R_0 ' (basic reproduction number). What does an R_0 of 3 mean for a disease?
2. Investigate one way that computer models were used during the COVID-19 pandemic to inform government decisions.



2.2.6 Identify how the body prevents the entry of pathogens and describe how it responds to pathogens that enter the body

The human body has three lines of defence against pathogens. The first line tries to stop pathogens getting in at all. If that fails, the second line responds quickly but non-specifically. If the pathogen is still a problem, the third line mounts a specific, powerful attack against that exact pathogen.

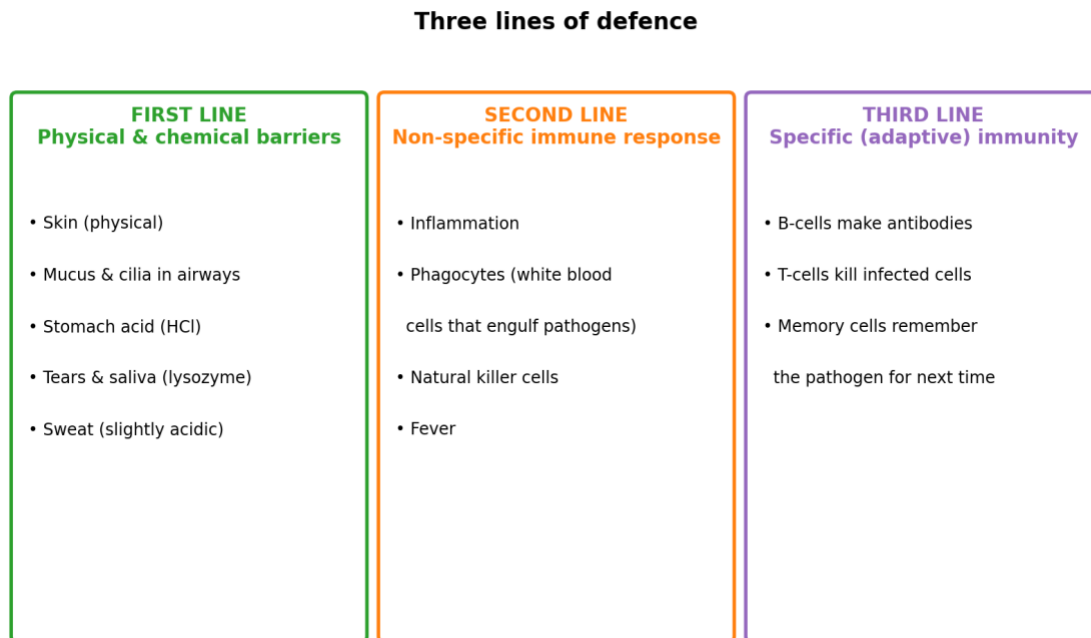


Figure 2.8 – Three lines of defence against pathogens.

First line: physical and chemical barriers

These barriers stop pathogens entering the body. They are not really part of the 'immune system' in the strict sense, but they are the reason most pathogens never cause an infection at all.

- Skin: a waterproof, multi-layered barrier.
- Mucus and cilia: line the airways, trap dust and pathogens, and sweep them out.
- Stomach acid: kills most bacteria in swallowed food.
- Tears and saliva: contain lysozyme, an enzyme that destroys bacterial cell walls.
- Sweat: slightly acidic, slowing bacterial growth on skin.
- Beneficial bacteria: live in the gut, mouth and skin, out-competing dangerous species.

Second line: non-specific immune response

If a pathogen gets past the first line, the body's innate (inborn) immune response begins. This response works the same way for every pathogen.

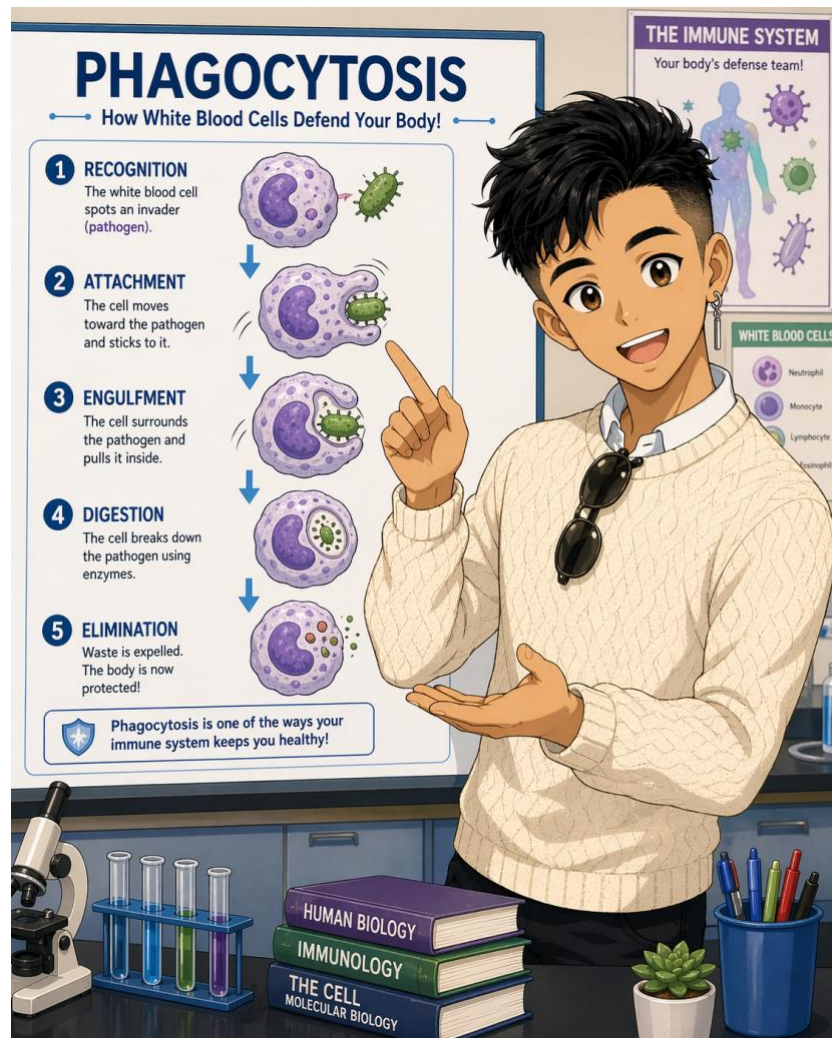
- Inflammation: damaged cells release chemicals such as histamine, widening blood vessels and making them leaky. The area becomes red, warm, swollen and sore – but it also fills with immune cells.
- Phagocytosis: white blood cells called phagocytes (e.g. macrophages and neutrophils) engulf and digest pathogens.

- Fever: the hypothalamus raises body temperature, which slows bacterial growth and speeds up immune reactions.
- Natural killer cells: a type of white blood cell that destroys virus-infected and cancerous body cells.

Third line: specific (adaptive) immune response

If the infection continues, lymphocytes (B-cells and T-cells) kick in. These are white blood cells that produce a specific response to that exact pathogen and remember it afterwards.

- B-cells produce antibodies. An antibody is a Y-shaped protein that binds only to a specific antigen (a part of the pathogen). Once bound, the pathogen is marked for destruction by other immune cells.
- T-cells (helper T-cells) activate other immune cells. T-cells (killer T-cells) destroy body cells that are infected with a virus.
- Memory cells are long-lived B- and T-cells that stay in the body after the infection, ready to respond much faster if the same pathogen is ever encountered again. This is why you only get chickenpox once.



Questions – 2.2.6

Comprehension Questions

1. List three physical or chemical barriers that prevent pathogens entering the body.
2. What is phagocytosis?
3. What is the role of a B-cell?
4. What is a memory cell?

Analysis Questions

1. Explain why swelling, redness and heat at a wound are actually a sign that the immune system is working.
2. Describe how the structure of an antibody relates to its function.
3. A person has their spleen removed (an organ containing many lymphocytes). Predict how this would affect their ability to fight infection.

Research / Extension Questions

1. Research one autoimmune disease (e.g. type 1 diabetes, rheumatoid arthritis, lupus). Explain how the immune system causes harm in this disease.
2. Investigate why HIV is such a dangerous virus, based on the cells it infects.

2.2.7 Outline how a vaccination stimulates the body to produce antibodies to fight infection

How a vaccine creates immunity

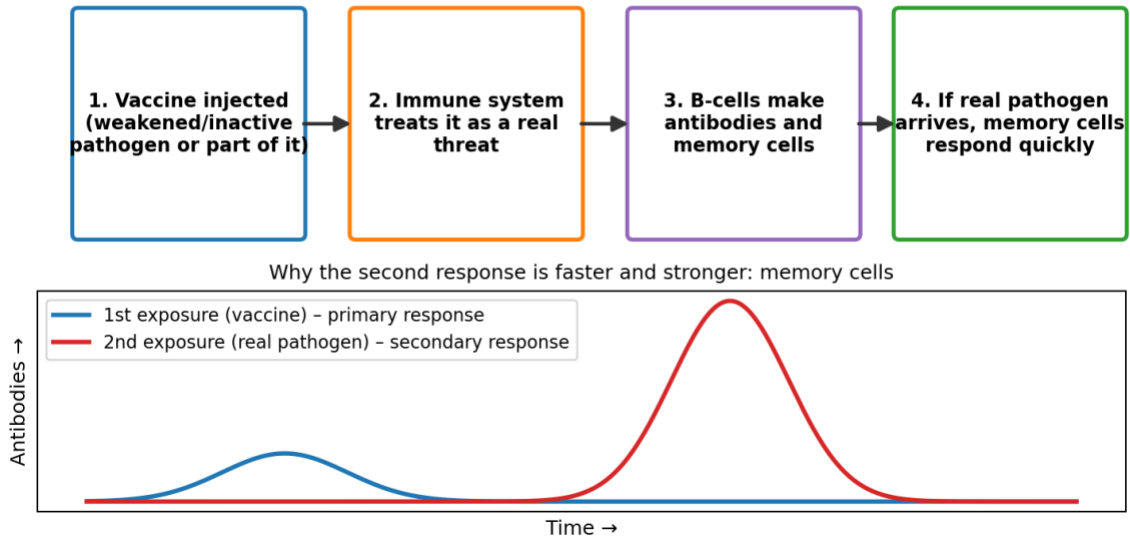


Figure 2.9 – How a vaccine creates immunity.

A vaccine is a preparation that contains a weakened, inactive or fragmentary form of a pathogen. When it is injected or swallowed, the immune system treats it as a real threat and mounts a full specific immune response, but without the person actually getting sick.

Step-by-step

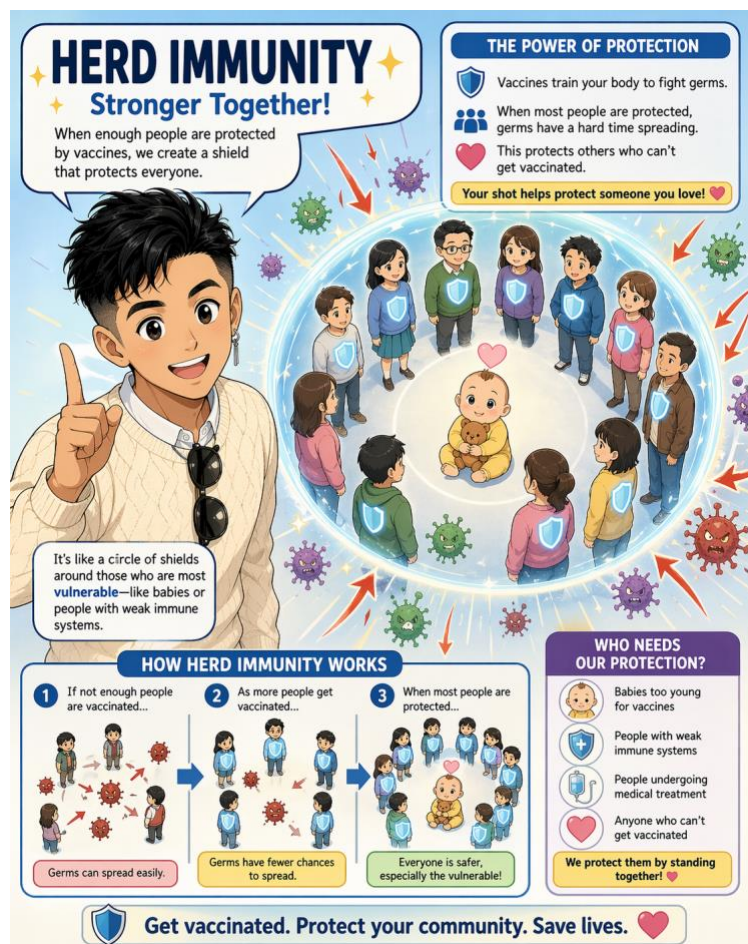
1. The vaccine is introduced into the body. It contains antigens from the pathogen, but the pathogen itself cannot cause disease.
2. The immune system recognises the antigens as 'non-self' and activates B-cells and T-cells.
3. B-cells divide rapidly, and some become plasma cells that release large numbers of antibodies specific to the vaccine antigen.
4. Some of the B-cells and T-cells become memory cells that persist in the body for years, and in some cases for life.
5. If the real pathogen ever enters the body later, memory cells recognise it immediately. The secondary response is much faster and larger than the primary response, so the person either does not get sick or has only mild symptoms.

Types of vaccines

Type	What is in it	Example
Live attenuated	Weakened version of the pathogen	Measles, mumps, rubella (MMR)
Inactivated (killed)	Whole pathogen that has been killed	Hepatitis A, rabies
Subunit / protein	Only a part (antigen) of the pathogen	Hepatitis B, HPV, Gardasil
mRNA	Genetic instructions for the body to build the antigen itself	Pfizer-BioNTech and Moderna COVID-19 vaccines
Viral vector	Harmless virus carrying the genes for an antigen	AstraZeneca COVID-19 vaccine

Herd immunity

When a very high proportion of a population is vaccinated, the pathogen struggles to find new hosts and cannot spread. This indirectly protects people who cannot be vaccinated – such as newborns, people undergoing chemotherapy, or those with severe allergies. This effect is called herd immunity. Different diseases require different vaccination rates to achieve it (measles, for example, requires around 95 % coverage).



Questions – 2.2.7

Comprehension Questions

1. What is a vaccine?
2. Name three types of vaccine.
3. What is a memory cell?
4. What is meant by 'herd immunity'?

Analysis Questions

1. Using Figure 2.9 (lower graph), explain why the second response to a pathogen is stronger than the first.
2. A friend says: 'I don't need the flu shot because I never get sick.' Using ideas from this section, give two reasons why they should still consider getting vaccinated.
3. Explain why vaccines do not work well against pathogens that change their antigens very often (such as the influenza virus).

Research / Extension Questions

1. Research the National Immunisation Program in Australia. List three vaccines that are recommended for Year 7 students.
2. Investigate how mRNA vaccines differ from traditional vaccines and why they were developed so quickly during the COVID-19 pandemic.

2.3 Disease control and prevention

Preventing disease is almost always easier, cheaper and more effective than treating it. In this section we look at how individuals, governments and communities reduce both non-infectious and infectious disease, including the long-established plant medicine knowledge of Aboriginal and Torres Strait Islander Peoples.

2.3.1 Describe ways to reduce the incidence of non-infectious diseases

Non-infectious diseases usually result from a mix of genetic, lifestyle and environmental factors. Genetic factors cannot easily be changed, but the other factors can. Around half of the risk of the most common chronic diseases in Australia is linked to modifiable lifestyle factors.

Individual action

- Eat a balanced diet – plenty of vegetables, fruit and wholegrains; limited saturated fat, added sugar and salt.
- Be physically active – the Australian guidelines recommend at least 60 minutes of moderate-to-vigorous activity each day for teenagers.
- Avoid tobacco – smoking is the single largest preventable cause of disease in Australia.
- Limit alcohol use – current Australian guidelines suggest no more than 10 standard drinks per week for adults, and none for people under 18.
- Protect skin from UV – hat, sunscreen SPF 50+, shade, and protective clothing reduce risk of melanoma and non-melanoma skin cancers.
- Manage stress and sleep – chronic stress and poor sleep raise the risk of cardiovascular and mental health conditions.
- Get regular health checks – early detection of high blood pressure, high cholesterol or pre-cancerous changes is far easier to manage than disease after it has developed.

Community and government action

- Taxation of tobacco and high-sugar drinks (to discourage use).
- Food labelling (e.g. the Health Star Rating) so consumers can make informed choices.
- Public education campaigns (e.g. LiveLighter, Slip-Slop-Slap-Seek-Slide).
- Built environment changes – bike lanes, parks and safe footpaths encourage activity.
- Screening programs – BreastScreen Australia, the National Bowel Cancer Screening Program, and cervical screening detect cancers early.
- Banning or restricting harmful substances (e.g. asbestos, lead in paint).

Disease	Main modifiable risk factor(s)	Main preventive actions
Type 2 diabetes	Obesity, inactivity, diet	Healthy diet, regular exercise, maintain healthy weight
Coronary heart disease	Smoking, high BP, high cholesterol, inactivity	Quit smoking, regular exercise, healthy diet, medication if needed
Lung cancer	Smoking, air pollution	Do not smoke, avoid secondhand smoke, reduce air pollution exposure
Melanoma	UV exposure	Sun protection (hat, sunscreen, shade), avoid solariums
Chronic respiratory disease (COPD)	Smoking, occupational dust	Quit smoking, workplace dust controls, vaccination against flu and pneumonia

Questions – 2.3.1

Comprehension Questions

1. List four lifestyle changes that can reduce the risk of chronic disease.
2. What does SPF 50+ protect against?
3. Name one Australian public health campaign.
4. Why is smoking described as the single largest preventable cause of disease in Australia?

Analysis Questions

1. Explain why some risk factors can be modified and others cannot.
2. The Australian government taxes cigarettes very heavily. Justify this policy using ideas from this section.
3. Suggest one reason why a healthy diet might be hard to achieve for some Australians, even if they know it is important.

Research / Extension Questions

1. Investigate the Health Star Rating system. What is on the front of an Australian food package, and how is the rating calculated?
2. Research one workplace safety regulation in Australia that exists to reduce non-infectious disease.

2.3.2 Assess ways to reduce the incidence and spread of infectious diseases

To reduce the spread of an infectious disease, we need to break the chain of infection (Figure 2.7). Different actions target different links in the chain.

Link in chain	Action to break it	Example
Pathogen	Kill pathogens or reduce their numbers	Antibiotics, disinfectants, cleaning surfaces, pasteurising milk
Reservoir	Treat infected people; control animal hosts	Isolating sick patients; controlling mosquito populations
Portal of exit	Cover coughs and sneezes; safely dispose of waste	Tissues, masks, proper sewage treatment
Transmission	Physical distance; barriers; vectors controlled	Masks, quarantine, mosquito nets, handwashing, safe water
Portal of entry	Cover wounds; safe food preparation	Bandages, gloves, food handling rules
Susceptible host	Improve immunity	Vaccination, good nutrition, rest

Layers of defence against infectious disease

No single intervention stops every infection. Effective control uses many layers of defence together.

6. Personal hygiene: handwashing with soap is still one of the most effective public-health measures ever developed.
7. Safe water and sanitation: clean drinking water and proper sewerage have prevented more disease than any medical treatment.
8. Safe food: refrigeration, cooking to safe temperatures, and food-handling rules.
9. Physical barriers: masks, gloves, mosquito nets.
10. Isolation and quarantine: separating sick or exposed people from the rest of the population.
11. Vector control: spraying, draining standing water, and wearing insect repellent.
12. Vaccination: the strongest long-term tool for most viral diseases.
13. Antibiotics and antivirals: used to treat infections, although resistance is an increasing problem.
14. Contact tracing and surveillance: identifying cases and their contacts so the chain can be broken early.

Assessing effectiveness

The effectiveness of any single measure depends on how the disease spreads. For example:

- Masks are very effective for droplet and airborne diseases (flu, COVID-19) but not for mosquito-borne diseases.
- Vaccination is very effective for viral diseases with stable antigens (measles) but less effective for rapidly mutating viruses (influenza).
- Antibiotics are effective against bacteria but useless against viruses, and overuse drives antibiotic resistance.

Questions – 2.3.2

Comprehension Questions

1. Name four ways an individual can reduce the spread of infectious disease.
2. What is meant by 'quarantine'?
3. Why are antibiotics useless against viral infections?
4. What is vector control?

Analysis Questions

1. For each link in the chain of infection, suggest one practical way it can be broken.
2. Assess which measure – vaccination, handwashing or masks – would be most effective against each of: malaria, influenza, gastroenteritis. Justify each choice.
3. Explain why overuse of antibiotics is dangerous for public health.

Research / Extension Questions

1. Research 'antibiotic resistance'. Explain how bacteria become resistant and why this is a problem for hospitals.
2. Investigate the Australian government's response to COVID-19. Identify three different public health measures that were used.



2.3.3 Investigate Aboriginal and/or Torres Strait Islander Peoples' use of plants to prevent or control disease

Aboriginal and Torres Strait Islander Peoples have lived continuously on this continent for at least 65,000 years. Over this time, different Nations developed extensive and sophisticated knowledge of local plants and their medicinal uses. This knowledge is held by specific Aboriginal and Torres Strait Islander communities and is passed down through generations. Out of respect, we should always name the specific Country and Peoples a plant and its use are associated with, rather than referring to all Aboriginal and Torres Strait Islander Peoples as one group.

A note on cultural knowledge

Much traditional medicinal knowledge – in particular, the exact methods of preparation and application – remains the cultural intellectual property of the relevant Aboriginal or Torres Strait Islander Nation. The examples in this booklet include only information that has been publicly shared by Traditional Custodians or is documented in published sources. Any deeper study of bush medicine should be led by Aboriginal and Torres Strait Islander voices.

Examples of traditional plant medicines by Country

Plant	Nation / Country	Traditional use	Scientific note
Melaleuca alternifolia (tea tree)	Bundjalung Peoples, coastal northern NSW	Crushed leaves applied to skin cuts, wounds, and infections; leaves inhaled for coughs and colds.	Essential oil contains terpinen-4-ol, shown in modern studies to have antimicrobial activity against bacteria and some fungi.
Pittosporum angustifolium (gumbi gumbi)	Yuwaalaraay, Kamilaroi and other Peoples of inland northern NSW and southern Queensland	Leaves boiled to make a tea drunk for colds, skin complaints and general wellness; sometimes applied to the skin.	Research has identified saponins and other compounds with anti-inflammatory and antimicrobial properties.
Eremophila spp. (emu bush)	Arrernte Peoples of Central Australia (around Mparntwe / Alice Springs), and other Central and Western Desert Nations	Crushed leaves applied to sores and cuts; infusions inhaled or drunk to treat colds and chest complaints.	Extracts have shown strong antibacterial activity in laboratory studies, including against MRSA.
Eucalyptus tetrodonta (stringybark) and other eucalypts	Yolŋu Peoples of northeast Arnhem Land (NT); also widely used by many other Nations across Australia	Leaves steamed or burned and the vapour inhaled to relieve colds, flu and blocked airways.	Eucalyptus oil contains 1,8-cineole, an active ingredient in many modern cold and chest preparations.
Centipeda cunninghamii (old man weed, 'gukwonderuk')	Yorta Yorta Peoples of the Murray–Goulburn region (VIC/NSW border)	Whole plant crushed and used as a wash or poultice for skin complaints, sores, and eye inflammation; sometimes taken internally for coughs.	Active compounds are being investigated for anti-inflammatory and wound-healing properties.
Santalum lanceolatum (native sandalwood) and other bush medicine plants	Western Desert Peoples (e.g. Pintupi, Ngaanyatjarra)	Leaves, bark and fruit used in preparations for skin rash, rheumatism and chest infections.	Essential oils contain antimicrobial terpenoids; compounds are still being studied.

Syzygium luehmannii / Backhousia citriodora (lilly pilly / lemon myrtle)	Peoples of the subtropical rainforests of the NSW north coast and southeast Queensland	Leaves and fruit used traditionally as antiseptics and as a source of vitamin C; lemon myrtle leaves used to freshen breath and aid digestion.	Lemon myrtle contains high levels of citral, which has strong antimicrobial activity; modern native-foods research continues.
Mangrove bark (Bruguiera spp.) and bush-medicine plants of saltwater Country	Bardi and Jawi Peoples (Dampier Peninsula, WA); also other saltwater Nations of northern Australia	Bark preparations used as washes for sores, boils and skin complaints.	Contains tannins and other antimicrobial compounds.

What was special about this knowledge?

- It is place-based: each Nation developed detailed knowledge of the plants of their own Country.
- It is holistic: 'medicine' often included food, song, ceremony and care for Country, not only a chemical effect.
- It is empirical: plants were selected and refined over many thousands of generations of observation – a very long experimental record.
- Many preparations have since been shown by modern chemistry and microbiology to contain genuinely active compounds. This includes antimicrobial essential oils, anti-inflammatory saponins and wound-healing flavonoids.

Respect and the way forward

Much of this knowledge was damaged by the disruption of colonisation, and some of it has been lost. Some has been exploited commercially without recognition of the Nations who developed it. When learning about or working with bush medicine, scientists and students should:

- Name the specific Nation / Country the knowledge comes from.
- Acknowledge that cultural knowledge belongs to the Traditional Custodians.
- Seek out sources authored or authorised by Aboriginal and Torres Strait Islander people.
- Support benefit-sharing when traditional knowledge is used in commercial products.

Questions – 2.3.3

Comprehension Questions

1. Why is it important to name the specific Nation or Country when discussing traditional plant medicines?
2. Name two plants used for skin complaints and the Peoples who used them.
3. What modern-day medical compound is related to eucalyptus use?
4. Which Peoples used gumbi gumbi, and what was it used for?

Analysis Questions

1. Explain why traditional plant medicines can be described as the result of a very long 'natural experiment'.
2. Laboratory studies have shown that tea tree oil kills bacteria. Does this mean Bundjalung knowledge of tea tree is now 'scientific'? Discuss.
3. Why do you think so many Aboriginal medicinal plants are sources of essential oils with antimicrobial activity?

Research / Extension Questions

1. Research ONE of the plants in the table in more depth. Identify the Nation(s) associated with it, the traditional preparation and use, at least one modern scientific study, and any current commercial products that use it.
2. Investigate what 'benefit-sharing' means in the context of traditional knowledge and modern pharmaceutical research, using one Australian or international example.

2.3.4 Analyse data about immunisation programs and the occurrence of infectious diseases to identify trends, patterns and relationships, and document conclusions in a written text

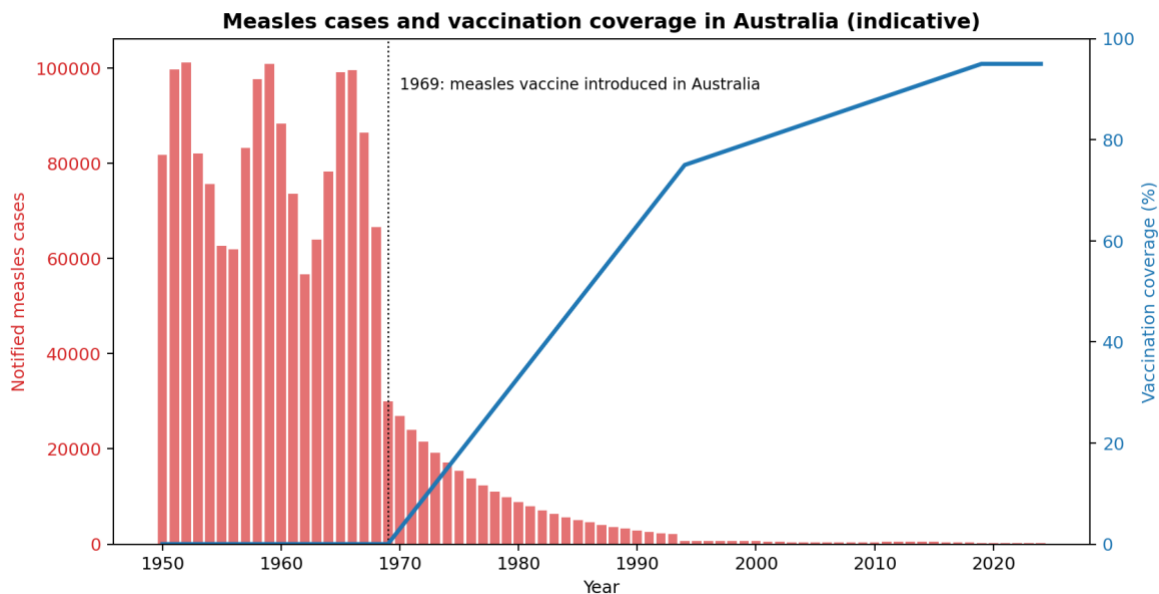


Figure 2.10 – Notified measles cases in Australia and estimated vaccination coverage (indicative only – values reconstructed for educational use).

In Australia and most developed countries, immunisation programs have transformed the pattern of infectious disease. The data above shows the general pattern for measles: very high numbers of cases before the vaccine, a sharp fall after vaccination began in 1969, and a further drop once two-dose vaccination became standard.

How to analyse disease-and-vaccination data

1. Identify the two variables shown on the graph (often time on the x-axis, and cases and/or coverage on the y-axis).
2. Describe the overall trend in cases before the vaccine was introduced.
3. Identify the point at which vaccination was introduced (often a line on the graph).
4. Describe what happens to cases after that point.
5. Describe the relationship between vaccination coverage and the number of cases.
6. Look for any time periods where cases rise briefly, as these often correspond to periods of lower coverage.

Data task – Analysing immunisation data

Instructions

Using Figure 2.10 or data you find yourself from the Australian Immunisation Handbook (health.gov.au) or the Australian Institute of Health and Welfare (aihw.gov.au), write a short scientific report (around 250–400 words) that:

1. Identifies the disease you are studying and the vaccine used to prevent it.
2. Describes the trend in cases before vaccination began.
3. Describes the trend in cases after vaccination began.
4. Describes the relationship between vaccination coverage and disease incidence.
5. Explains the pattern using ideas from 2.2.7 (how vaccines work) and 2.3.2 (breaking the chain of infection).
6. Draws a clear conclusion about the effectiveness of the immunisation program.

Scientific writing tips

- Use the data. Quote numbers or percentages from the graph in your text.
- Use the past tense to describe trends ('cases fell from ... to ...').
- Use the word 'correlated' carefully. Say 'as coverage increased, cases decreased' rather than 'because of vaccination, cases fell', unless the data clearly supports causation.
- Reference your data source at the end.

Example pattern to look for in real data

Disease	Before vaccination (Australia)	Now (Australia)
Smallpox	Major epidemics in 18th–19th century	Eradicated globally (1980)
Polio	Hundreds of cases a year, many paralytic	No locally acquired cases since 2000
Measles	Tens of thousands of cases per year	Usually < 200 cases per year, mostly linked to overseas travel
Diphtheria	A leading killer of children	Fewer than 10 cases per year
Whooping cough	Thousands of cases per year, many infant deaths	Cases now mostly limited to outbreaks in unvaccinated communities

Questions – 2.3.4

Comprehension Questions

1. What year was the measles vaccine introduced in Australia?
2. Roughly how high was vaccination coverage by the year 2000?
3. What overall relationship exists between vaccination coverage and number of cases?
4. Name one disease that has been eradicated globally by vaccination.

Analysis Questions

1. Using Figure 2.10, describe what happened to measles cases in the ten years after vaccination began.
2. Explain why measles cases did not reach zero immediately after vaccination was introduced.
3. Some parents choose not to vaccinate their children. Using the data from this section, predict what would happen if vaccination coverage in Australia fell to 50 %. Justify your answer.

Research / Extension Questions

1. Research the World Health Organization's current list of vaccine-preventable diseases. Which one is closest to being eradicated next?
2. Investigate one country where vaccination coverage has recently dropped, and describe the impact on disease cases.



2.4 Disease – in Context

2.4.1 Investigate technological advances developed in Australia to address disease, disorders or physical trauma in the human body

Australian scientists and engineers have been responsible for a remarkable number of medical technologies that are now used around the world. In this section you will investigate some of these innovations and then carry out a deeper investigation of one of your choice.

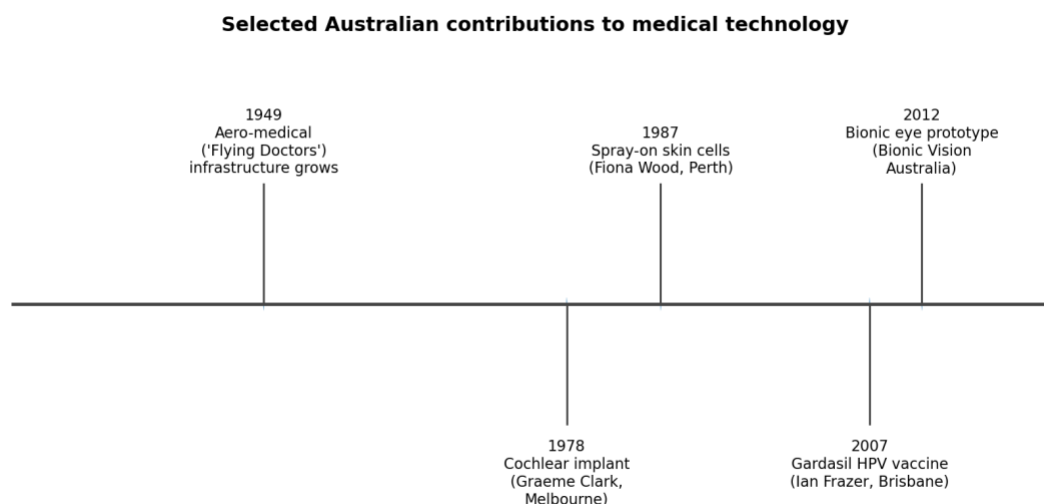


Figure 2.11 – Some notable Australian contributions to medical technology.

Case file: five Australian medical innovations

Innovation	Developer	Year	What it does / Impact
Cochlear implant ('bionic ear')	Professor Graeme Clark, University of Melbourne	1978 (first recipient)	An electronic device that bypasses damaged hair cells in the inner ear by directly stimulating the auditory nerve, restoring a sense of hearing to profoundly deaf people. Now used by over 700,000 people worldwide.
Spray-on skin cells (ReCell®)	Professor Fiona Wood and Marie Stoner, Perth	1987 onwards; used widely after the 2002 Bali bombings	A small piece of the patient's own healthy skin is processed into a suspension of skin cells that is sprayed onto burn wounds. This dramatically reduces healing time compared with traditional

			skin grafts and results in better scarring outcomes.
Gardasil (HPV vaccine)	Professor Ian Frazer and Dr Jian Zhou, University of Queensland	First approved 2006 (subunit vaccine)	A vaccine against human papillomavirus (HPV), the virus that causes almost all cases of cervical cancer. Australia became the first country to introduce a national HPV vaccination program, and cervical cancer cases in young Australian women have fallen dramatically.
Cervical cancer screening: the 'Pap smear' improvements	Professor George Papanicolaou's test, refined by Australian pathology services	1960s onwards	Though the original test was developed overseas, Australia built one of the most effective national screening programs in the world, now using HPV testing.
Bionic eye prototypes	Bionic Vision Australia consortium	2012 (first prototype implanted)	A device implanted behind the retina that receives signals from a camera mounted on glasses and stimulates the remaining retinal cells. Aimed at restoring partial vision in people with degenerative eye conditions such as retinitis pigmentosa.

Other significant Australian contributions

- Royal Flying Doctor Service (RFDS): founded in 1928, now delivering more than 300 000 patient contacts per year across remote Australia.
- Ultrasound imaging in obstetrics – pioneering work in Sydney in the 1950s–60s helped establish obstetric ultrasound as a routine safe test.
- Discovery of *Helicobacter pylori* as the cause of stomach ulcers – Robin Warren and Barry Marshall (Perth), 2005 Nobel Prize in Medicine.
- Relenza (zanamivir): an antiviral drug for influenza, developed by CSIRO and Biota in the 1990s.
- Extracorporeal membrane oxygenation (ECMO) advances: Australian hospitals have been at the forefront of critical care innovation.

Research task – Australian medical innovation profile

Task

Choose ONE Australian medical technology, treatment, or invention from the list above (or another approved by your teacher). Research it in depth and present your findings as one of: a one-page profile document, a 3–5 minute oral presentation, a short infographic, or a short video.

Your profile must address all of the following

1. What medical problem does this technology, disease or trauma address? Describe the problem clearly.
2. Who developed it, and where in Australia?
3. How does it work? Include a labelled diagram.
4. What impact has it had? Use statistics if possible (e.g. number of patients, reduction in cases or deaths, countries where it is used).
5. What limitations does it have, and what are researchers working on next?
6. Does the innovation involve Aboriginal and/or Torres Strait Islander knowledge, partnership or benefit? (Honest answer required – not all do.)

Sources

Use at least three reliable sources, including at least one scientific article or reputable Australian organisation (e.g. CSIRO, NHMRC, Australian Institute of Health and Welfare, or a university research page). List your sources in-text and at the end.

Marking rubric (for self-check)

- Accuracy of the science (/10)
- Clarity of writing / presentation (/5)
- Quality and variety of sources (/5)
- Reflection on impact and limitations (/5)

Questions – 2.4.1

Comprehension Questions

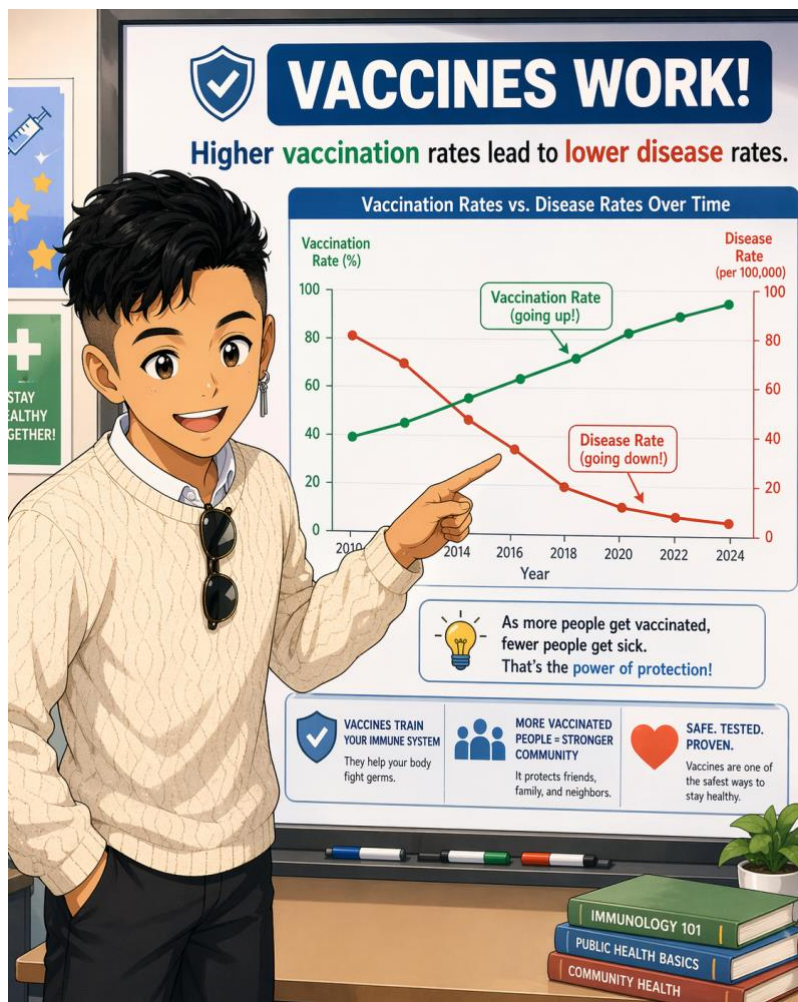
1. Who developed the cochlear implant and where?
2. What does spray-on skin do that traditional skin grafts cannot?
3. What does HPV stand for, and why is the Gardasil vaccine important?
4. What does the Royal Flying Doctor Service do?

Analysis Questions

1. Explain why the cochlear implant is a treatment for deafness, rather than a cure for it.
2. The Gardasil vaccine was publicly funded in Australia before many other countries. Suggest two reasons why.
3. Describe one ethical issue that might come up when developing new medical technologies such as a bionic eye.

Research / Extension Questions

1. Research the work of Barry Marshall and Robin Warren on *Helicobacter pylori*. Why is their discovery described as one of the most important in modern Australian medical research?
2. Investigate one Australian medical technology currently in development (not yet on the market). Describe what problem it aims to solve.



Glossary

Adaptive (specific) immunity – Immune response involving B- and T-cells that targets a specific pathogen and remembers it.

Antibody – A Y-shaped protein made by B-cells that binds specifically to an antigen.

Antigen – A molecule (often on a pathogen) that the immune system recognises as non-self.

Antibiotic – A medicine that kills bacteria or slows their growth. Does not work on viruses.

Bacteria – Single-celled living organisms; some cause disease.

B-cell – A type of white blood cell that produces antibodies.

Chronic disease – A long-lasting, non-infectious disease.

Cilia – Tiny hair-like projections on cells lining the airways that sweep mucus and trapped particles out.

Effector – A muscle or gland that carries out the body's response.

Endemic – A disease constantly present in a population at a low level.

Endocrine system – The system of glands that release hormones into the bloodstream.

Epidemic – A sudden, larger-than-expected increase in cases of a disease in a region.

Feedback loop – A system that uses the result of a change to control that same change.

Herd immunity – Indirect protection of a population when a high proportion are vaccinated or immune.

Homeostasis – The maintenance of stable internal conditions in an organism.

Hormone – A chemical messenger released into the blood by an endocrine gland.

Incidence – The number of new cases of a disease in a period of time.

Infectious disease – A disease caused by a pathogen that can be spread between hosts.

Innate (non-specific) immunity – The body's general, rapid response to any pathogen (inflammation, phagocytes, fever).

Lymphocyte – A type of white blood cell (B-cell or T-cell) involved in specific immunity.

Memory cell – A long-lived B- or T-cell that remembers a pathogen and responds quickly on re-exposure.

Nervous system – The system of neurons that carries fast electrical signals between the brain, spinal cord and the rest of the body.

Non-infectious disease – A disease not caused by a pathogen; cannot be passed between people.

Pandemic – An epidemic that spreads across many countries or continents.

Pathogen – An organism or virus that can cause disease.

Phagocyte – A type of white blood cell that engulfs and digests pathogens.

Prevalence – The total number of existing cases of a disease in a population at one time.

Receptor – A structure (cell or molecule) that detects a stimulus.

Reflex – A rapid, automatic response to a stimulus that does not require conscious thought.

Stimulus – A detectable change in the environment.

T-cell – A type of white blood cell that helps coordinate the immune response or kills infected cells.

Vaccine – A preparation that stimulates the immune system to develop immunity against a specific pathogen.

Vector – An organism (often an insect) that transmits a pathogen from one host to another.

Virus – A non-cellular infectious agent that requires a host cell to reproduce.

Syllabus self-check

Tick each dot point as you complete it. Use your book, your notes and this booklet's question boxes to check your understanding.

Dot point	Description	Confidence (✓ / ? / X)
2.1.1	Identify the importance of maintaining stable internal conditions in the body	
2.1.2	Investigate examples of an organism's observable response to a stimulus	
2.1.3	Identify the role of feedback loops in maintaining homeostasis	
2.1.4	Compare and contrast the responses of the nervous and endocrine systems	
2.1.5	Describe how the nervous and endocrine systems coordinate the body's response to stimuli	
2.2.1	Distinguish between infectious and non-infectious diseases	
2.2.2	Identify causes of non-infectious and infectious diseases	
2.2.3	Compare the features and incidences of epidemics, endemics and pandemics	
2.2.4	Investigate data relating to a common non-infectious disease affecting Australians today	
2.2.5	Use modelling to investigate how infectious diseases can be spread	
2.2.6	Identify how the body prevents the entry of pathogens and describe how it responds to pathogens that enter the body	
2.2.7	Outline how a vaccination stimulates the body to produce antibodies to fight infection	
2.3.1	Describe ways to reduce the incidence of non-infectious diseases	
2.3.2	Assess ways to reduce the incidence and spread of infectious diseases	
2.3.3	Investigate Aboriginal and/or Torres Strait Islander Peoples' use of plants to prevent or control disease	
2.3.4	Analyse data about immunisation programs and the occurrence of infectious diseases, and document conclusions	
2.4.1	Investigate technological advances developed in Australia to address disease, disorders or physical trauma	